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Network Diagrams and Activity on Node Analysis (AON)

MGMT-3105

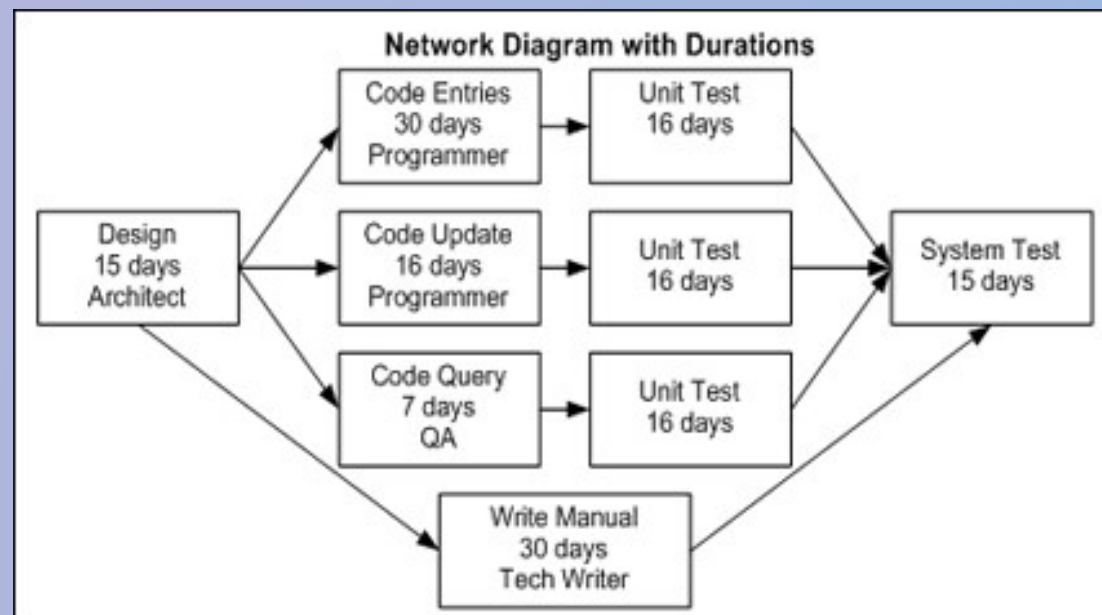
Lecture 6

Project Time Management

- **Project Time Management** includes all the process that are needed to manage the timely completion of the project.
 1. Plan Schedule Management
 - 2. Define Activities**
 - 3. Sequence the activities**
 4. Estimate activity resources
 - 5. Estimate activity durations**
 6. Develop schedule
 7. Control schedule.
- Today we are covering only the processes in **bold Red**

Define Activities Process

- First you need to generate the **Work Breakdown Structure (WBS)** for the project as demonstrated. This is done by identifying all the activities that need to be completed to meet the project deliverables.
- From the WBS, you need to generate the **Network Activity Diagram** for the project – commonly referred to as the Network Diagram.

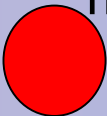


WBS Review

What is the definition of WBS?

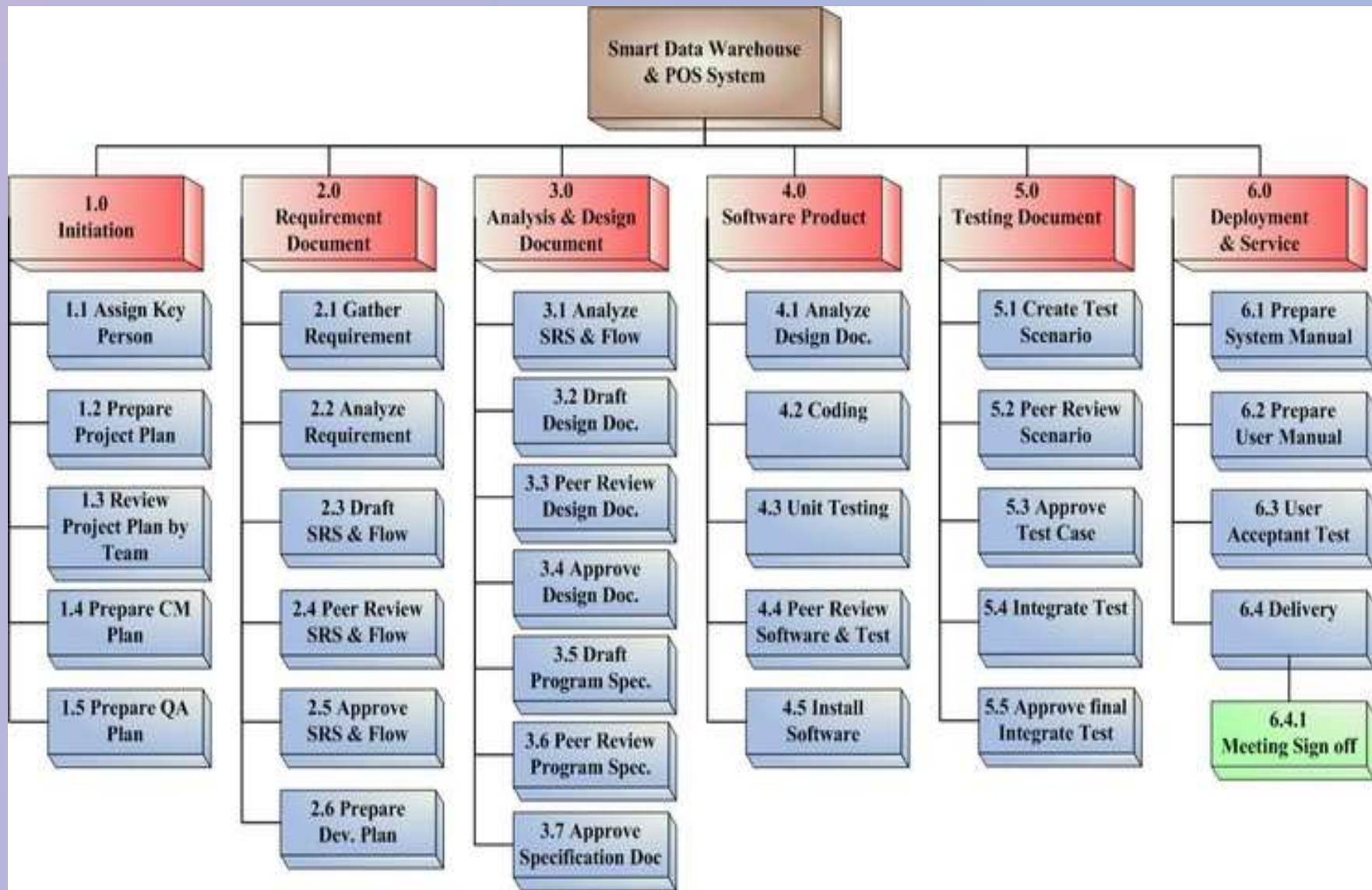
A work breakdown structure (*WBS*) is a deliverable oriented grouping of project components that organizes and defines the total scope of the project.

- ❖ The **WBS** is the **project scope** broken down into smaller manageable pieces or tasks. If all the tasks are completed, then the sum of those tasks will equal the project scope.
- ❖ **Level 1** of a WBS is the scope broken down in **broad categories**.
- ❖ **Level 2** is each of these broad categories broken down into smaller more manageable tasks or smaller categories (**work packages**). These tasks take between 8 and 80 hours to complete for the average project. You do not want to break tasks down below 8 hours as it becomes micromanaging at that point. Tasks more than 80 hours are too broad to manage properly.



<u>Sample WBS for Project Management</u>						
	1 Fabrication and Assembly of components for tool rest (Project 275)					
(Level 1)	1 Material					
	1.1	Material lists prepared from drawings				
(Level 2)	1.3	Material Ordered				
	1.5	Material inspected upon delivery				
	1.7	Prepare material for fabrication				
(Level 3)		1.7.1	Stock squared for base plates			
		1.7.3	Round stock sized for turning			
	3 Equipment					
	3.1	CNC 1 and 2 booked for required fabrication time				
	3.3	Lathes 1, 2 and 3 booked for required fabrication time				
	5 Manpower					
	5.1	Manpower availability with Fabrication Manager determined				
		5.1.1	Create / Post schedule for Fabrication operators			
	5.3	Manpower availability with Assembly Manager determined				
		5.3.1	Created / Posted schedule for assembly operators			
	5.5	Manpower availability with Quality Manager determined				
		5.5.1	Created / Posted schedule for Quality technician			
	7 Fabrication					
	9 Quality Inspections					
	11 Assembly					
	13 Quality Inspections					
	15 Customer Inspection					
	17 Packaging					
	19 Shipping					

WBS Example



The Rules for creating Network Activity Diagrams

1. Some determination of **activity precedence ordering** must be done prior to creating the network.
2. Network diagrams usually flow from **left to right**.
3. An activity cannot begin until all preceding **connected activities** have been completed.
4. **Arrows** on networks indicate **precedence** and **logical flow**. Arrows can cross over each other, although it is helpful for clarity's sake to limit this effect when possible.
5. Each activity should have a **unique identifier** associated with it (number, letter, code, etc.).
6. Looping, or recycling through activities, is **not permitted**.
7. Although not required, it is **common** to start a project from a **single beginning node**.
8. A single node point also is typically used as a **project end indicator**.



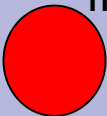
Key Scheduling Terminology - simplified

- **Activity:**
- **Scope:** The work content and products of a project.
- **Work Breakdown Structure (WBS):** A task-oriented grouping of activities that organizes, defines, and graphically displays the total work to be accomplished in order to achieve the final objectives (deliverables) of the project.
- **Work Package:** A deliverable at the lowest level of the work breakdown structure.
- **Project Network Diagram:** Any schematic display of the logical relationships of project activities.
- **Path:** A sequence of activities defined by the project network logic.
- **Early Start Date (ES):** The earliest possible date on which the uncompleted portions of an activity can be started.
- **Late Start Date (LS):** The latest possible date that an activity may begin without delaying a specified milestone.



Key Scheduling Terminology – simplified (cont.)

- **Early Finish**
- **Late Finish**
- **Forward Pass** – Network calculations that determine the **earliest** start/finish time for each activity.
- **Backward Pass** – Calculation of **late start** and **late finish times** for all uncompleted network activities.
- **Event** – A point in time when an activity is either **started** or **completed**.
- **Node** – One or more defining points of a network.
- **Predecessors** – Those activities that must be completed **prior** to an activity in the network.
- **Successors** – Activities that **cannot be started** until previous activities have been completed.
- **Merge Activity** – An activity with **2 or more immediate predecessor activities** flowing into it.



Key Scheduling Terminology – simplified (cont.)

- **Burst Activity:** An activity with **2 or more immediate successor activities** flowing out from it.
- **Float (Slack):** The **amount of time** an activity **may be delayed** from its **early start** without delaying the finish of the project.
- **Critical Path:** The path through the project network with the longest duration.
- **Critical Path Method:** A network analysis technique used to determine which sequence of activities (which path) has the least amount of scheduling flexibility and therefore will most likely determine when the project can be completed.

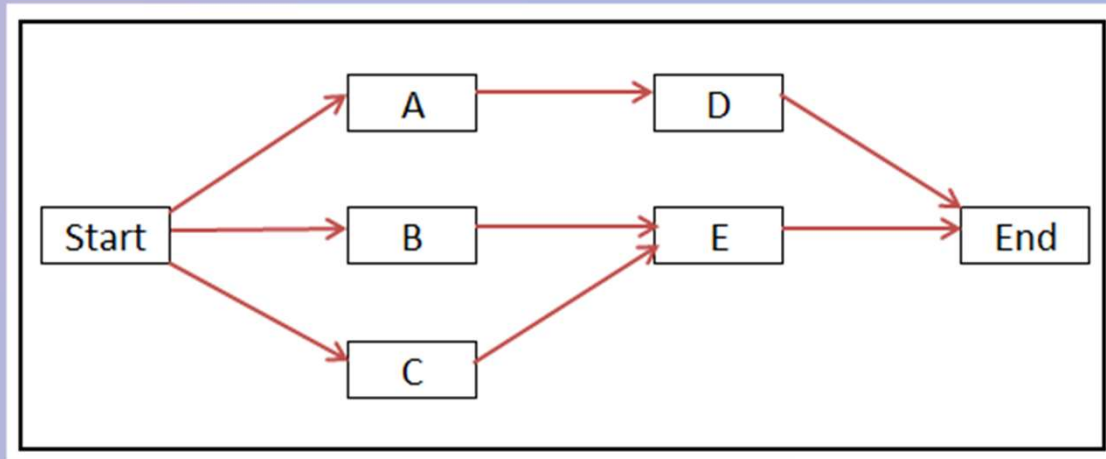


Key Scheduling Terminology – simplified (cont.)

- ***Resource Limited Schedule:*** A project schedule whose start and finish dates reflect **expected resource availability.**
- ***Program Evaluation and Review Tool (PERT):***
An event and probability-based network analysis system generally used in projects where activities and their durations are difficult to define.



Labels for Activity Nodes



ES	DUR	EF
Activity A		
LS	FLT	LF

ES Earliest Start

EF Earliest Finish

LS Latest Start

LF Latest Finish

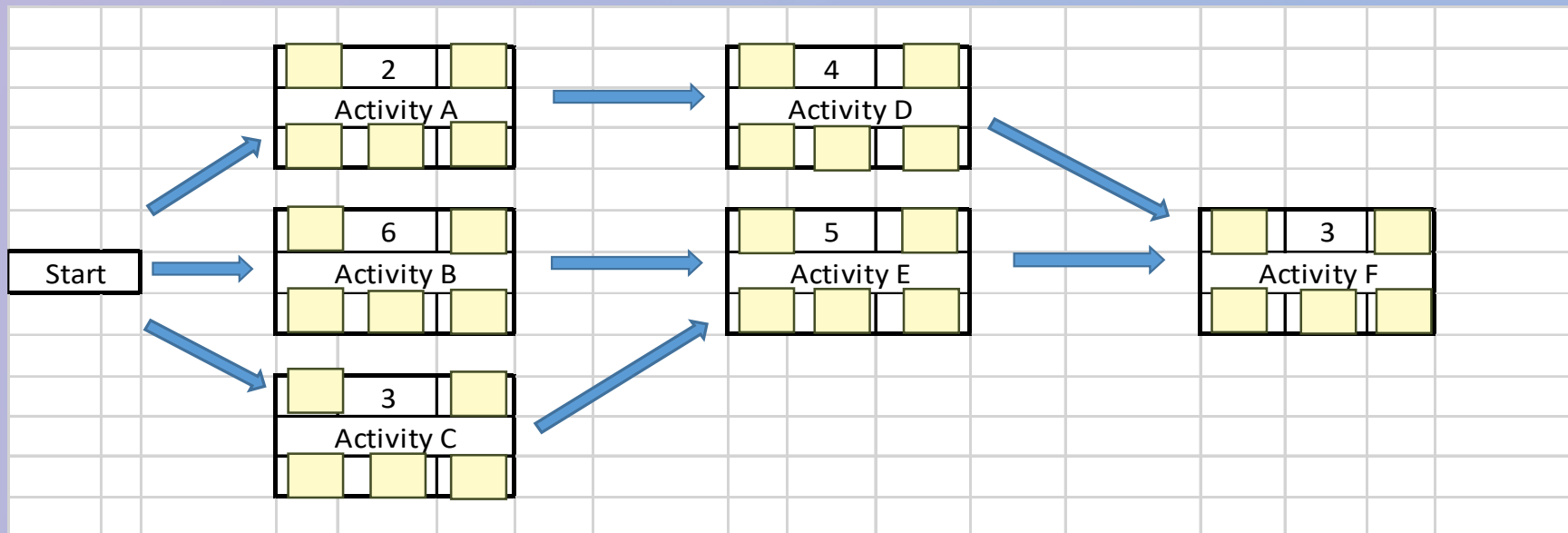
DUR Duration of Activity

FLT Float



Defining the numbers

1) Do the Forward Pass first to determine ES and EF



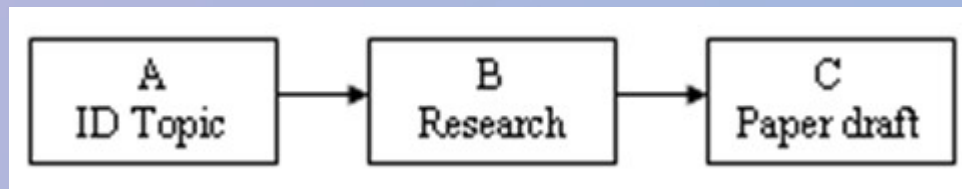
2) Do a Backwards Pass next to determine LF and LS

3) Subtract ES from LS to determine FLOAT

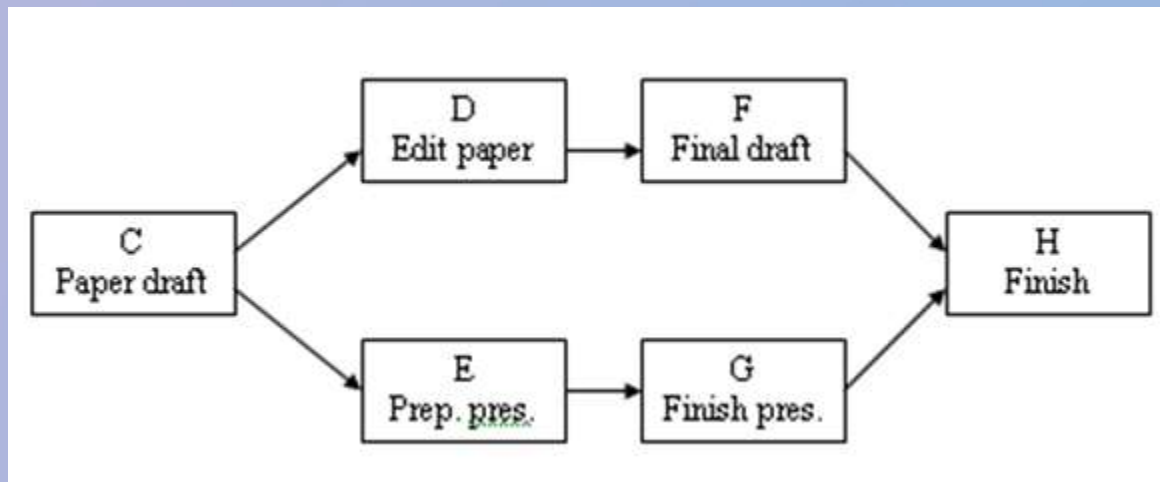


Series/Parallel Activities

Series

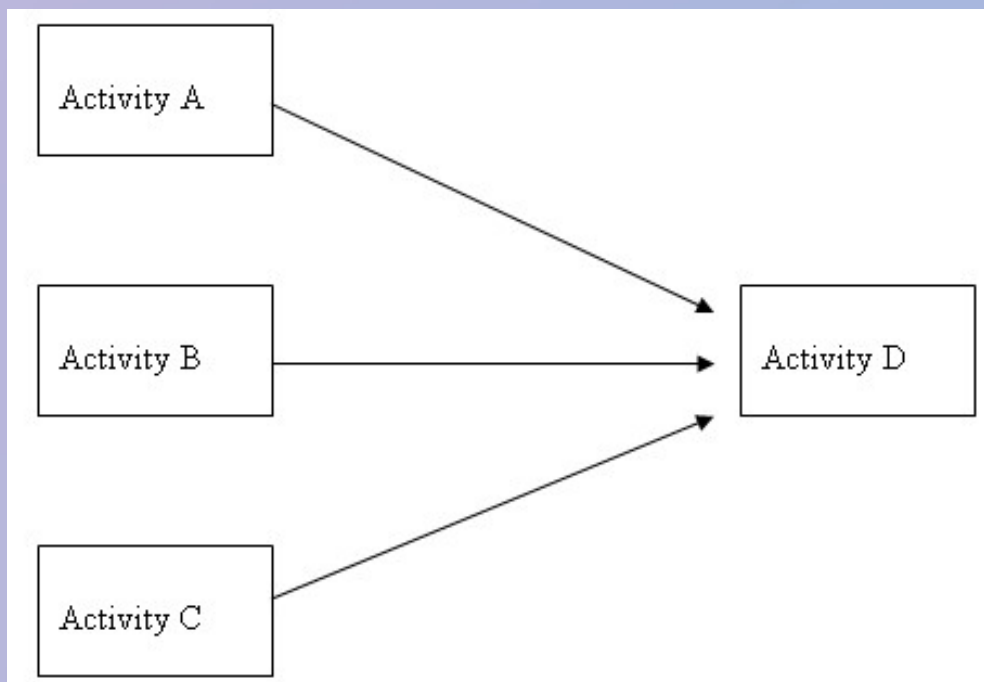


Parallel

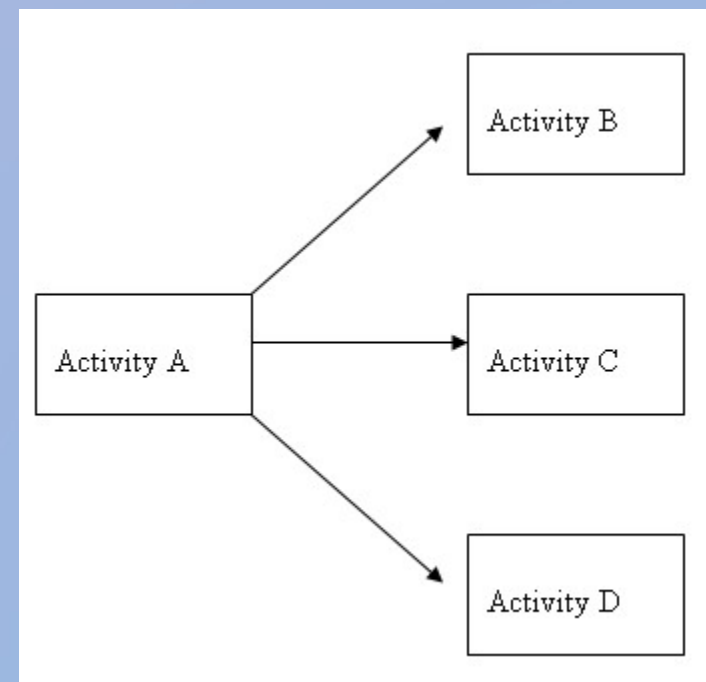


Activities Merge / Burst

Merge



Burst



Creating a Project Network Activity Diagram - Example

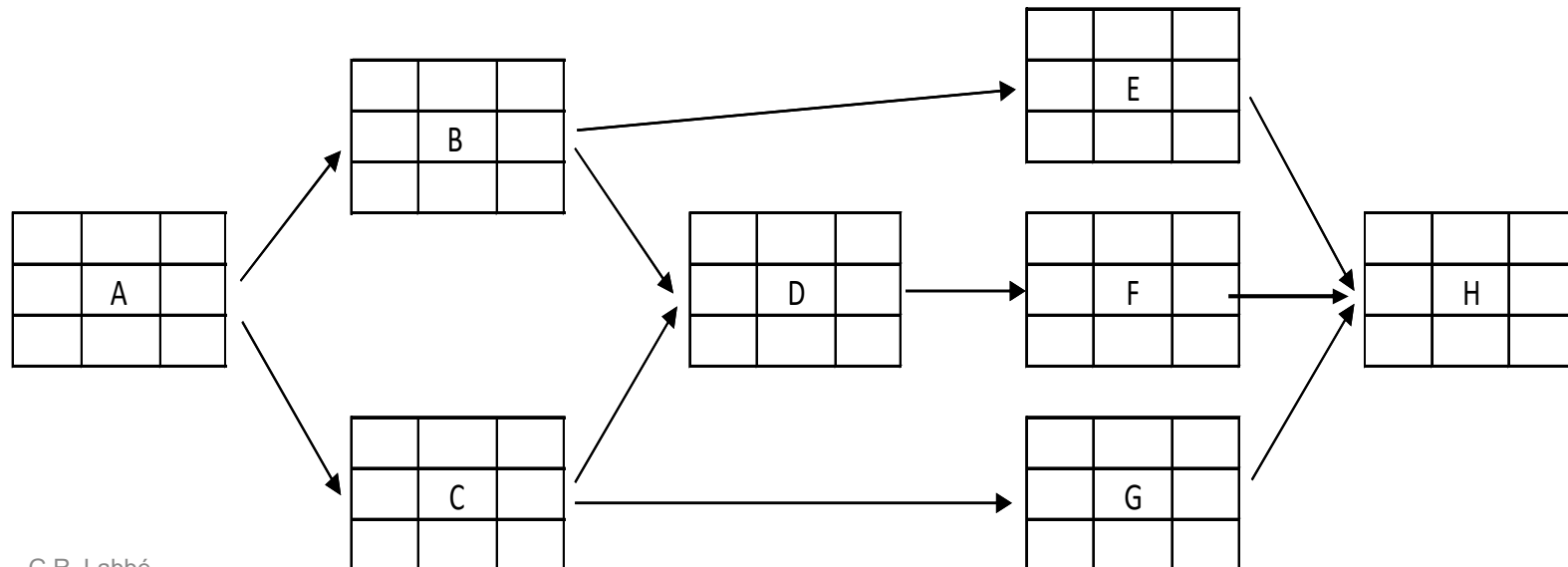
Information for Network Construction		
Name: Project Delta		
Activity	Description	Predecessors
A	Contract signing	None
B	Questionnaire design	A
C	Target market ID	A
D	Survey sample	B, C
E	Develop presentation	B
F	Analyze results	D
G	Demographic analysis	C
H	Presentation to client	E, F, G



Activity on Node Network

Information for Network Construction
Name: Project Delta

Activity	Description	Predecessors
A	Contract signing	None
B	Questionnaire design	A
C	Target market ID	A
D	Survey sample	B, C
E	Develop presentation	B
F	Analyze results	D
G	Demographic analysis	C
H	Presentation to client	E, F, G



Constructing the Critical Path

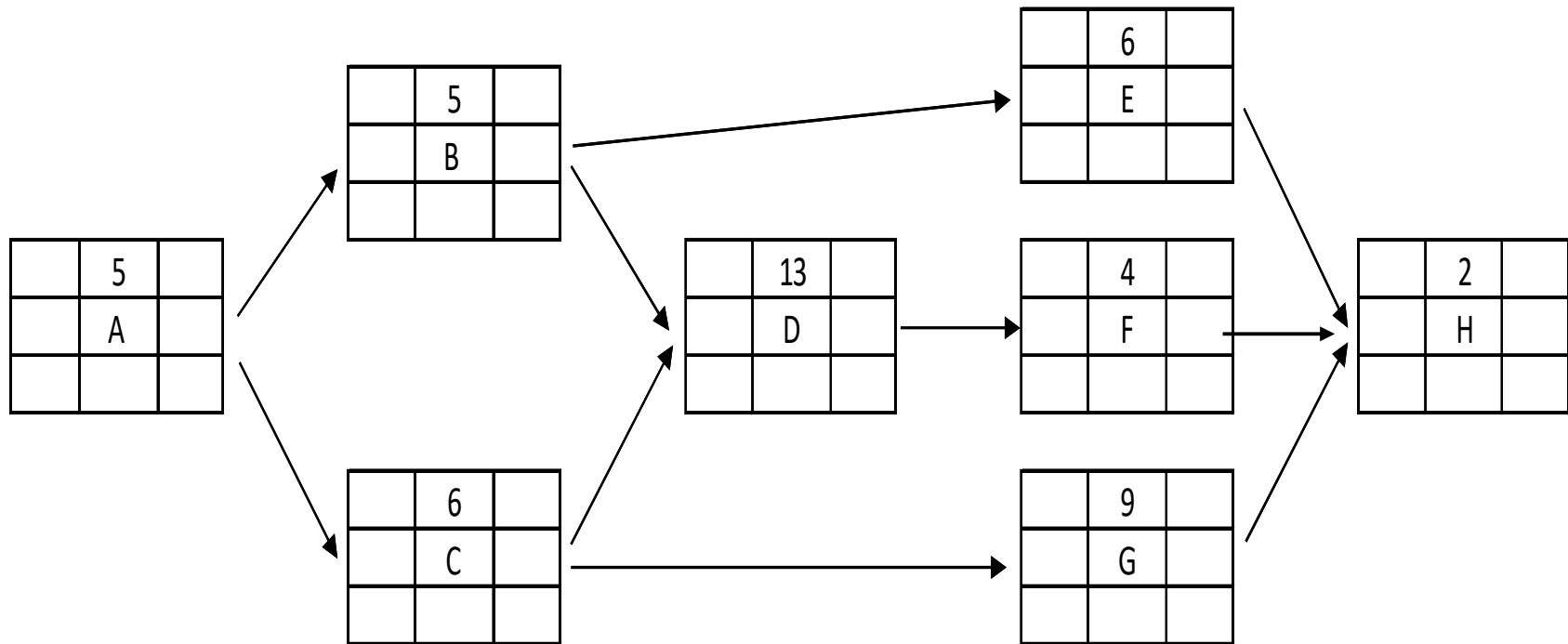
INFORMATION FOR PROJECT DELTA

<u>Activity</u>	<u>Description</u>	<u>Predecessors</u>	<u>Estimated Duration</u>
A	Contract signing	None	5
B	Questionnaire design	A	5
C	Target market ID	A	6
D	Survey sample	B, C	13
E	Develop presentation	B	6
F	Analyze results	D	4
G	Demographic analysis	C	9
H	Presentation to client	E, F, G	2

Project Activity Network Showing Project Durations

INFORMATION FOR PROJECT DELTA

Activity	Description	Predecessors	Estimated Duration
A	Contract signing	None	5
B	Questionnaire design	A	5
C	Target market ID	A	6
D	Survey sample	B, C	13
E	Develop presentation	B	6
F	Analyze results	D	4
G	Demographic analysis	C	9
H	Presentation to client	E, F, G	2

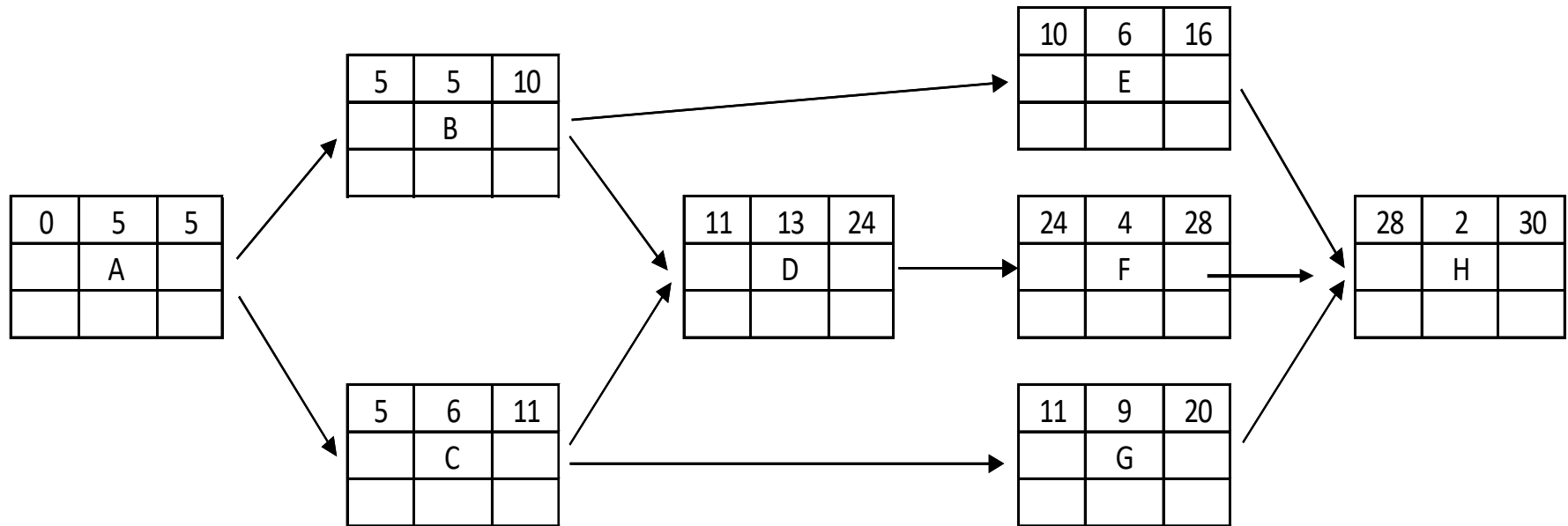


Rules when using the Forward Pass

1. Add all activity times along each path as we move through the network ($ES + \text{Duration} = EF$)
2. Carry the EF time to the activity nodes immediately ahead of the previous completed node. That EF becomes the **ES** of the next node, unless the next node is a merge point.
3. At a merge point, the **largest** previous EF becomes the ES for that node



Activity Network with Forward Pass

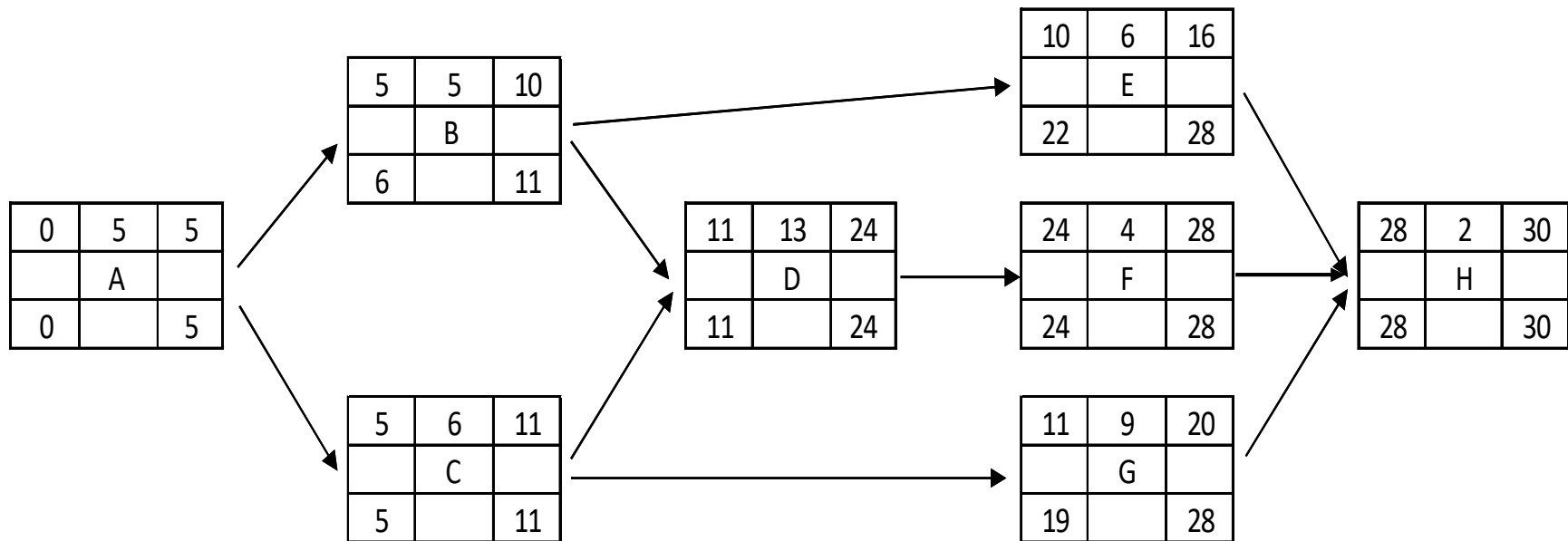


Rules when using the Backward Pass

1. Subtract activity times along each path as you move through the network ($LF - Dur = LS$)
2. Carry back the LS time to the activity nodes immediately before the next node. That **LS becomes the LF of the previous node**, unless the previous node is a burst point.
3. In the case of a burst point, the **smallest** following LS becomes the LF for that node.

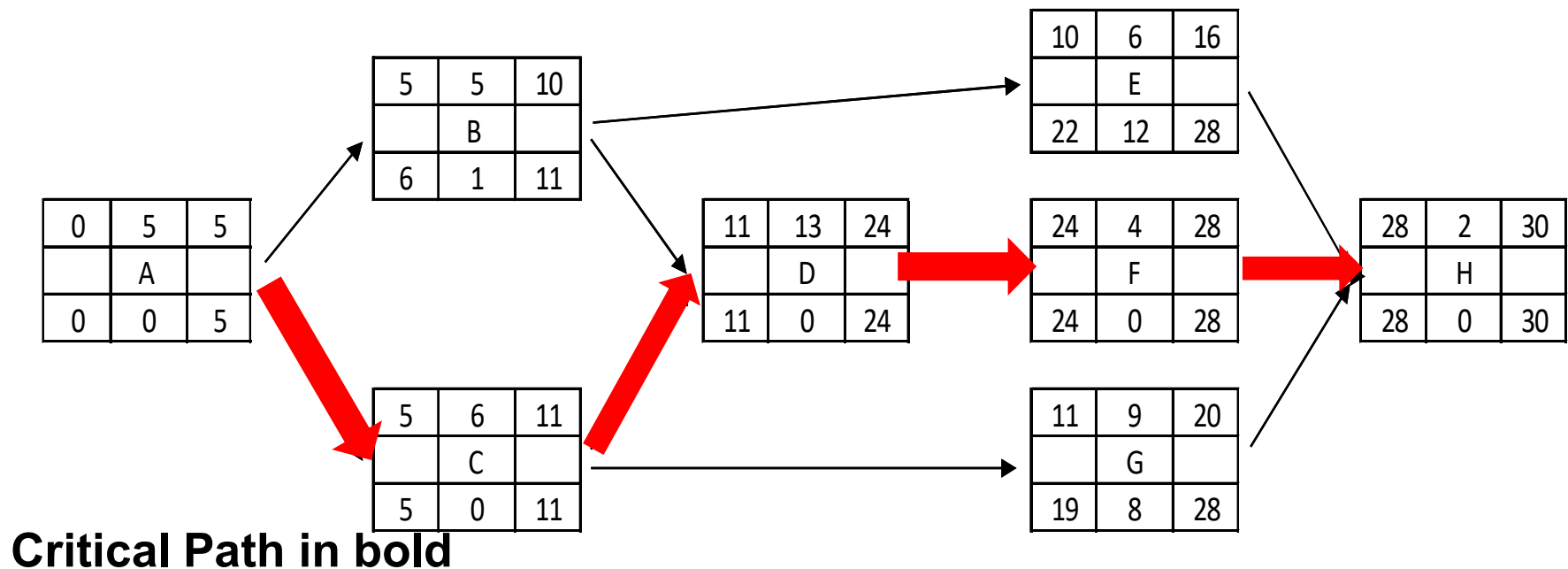


Activity Network with Backward Pass



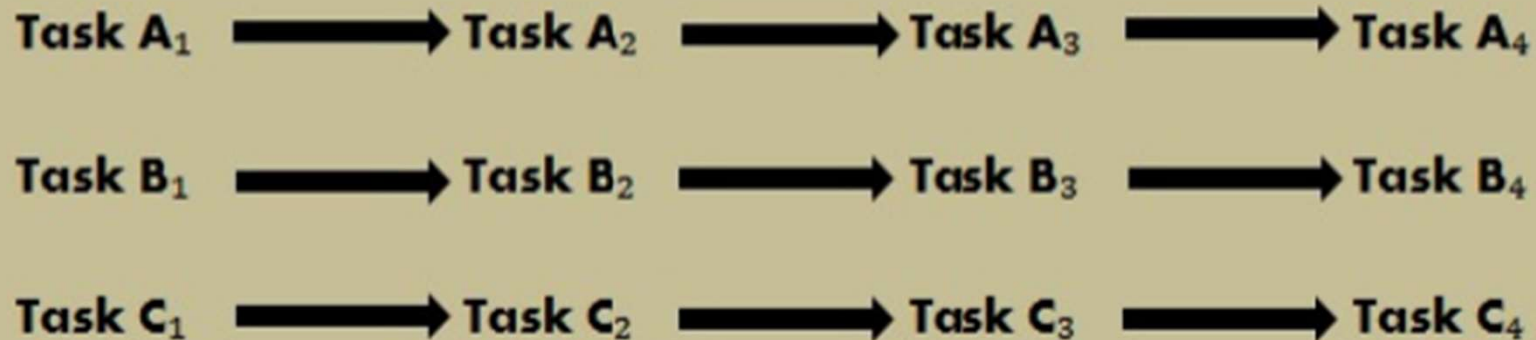
Completed Network Activity Diagram with Critical Path and Slack Times Calculated

$LS - ES$ or $LF - EF =$ the Float. Activities that have a zero float cannot be delayed without delaying the entire project. These tasks together form the critical path.



Network Activity Diagram example that Illustrates Laddering technique

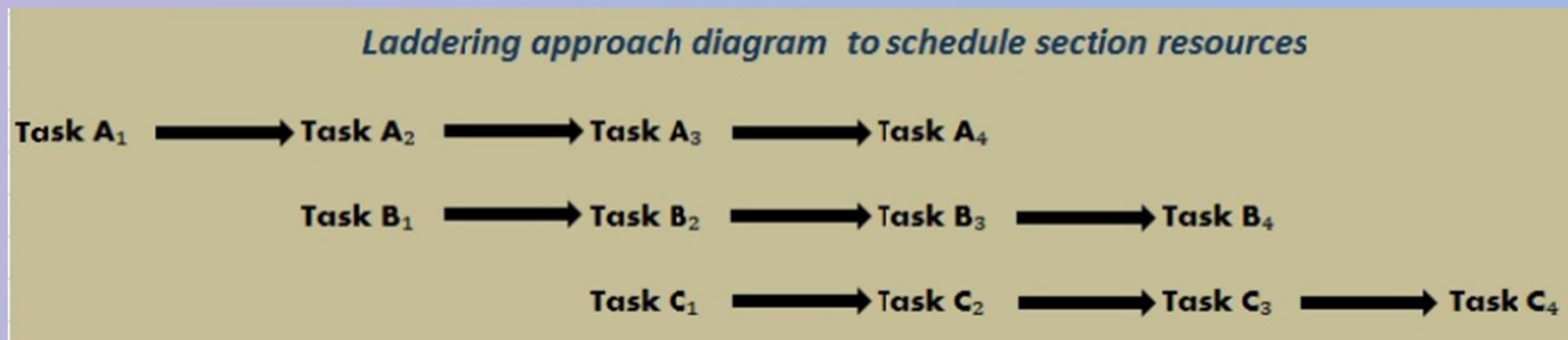
AON(PDM) diagram resource independent



If Tasks A₁, B₁ and C₁ **do not need the same resources** then they could run simultaneously.



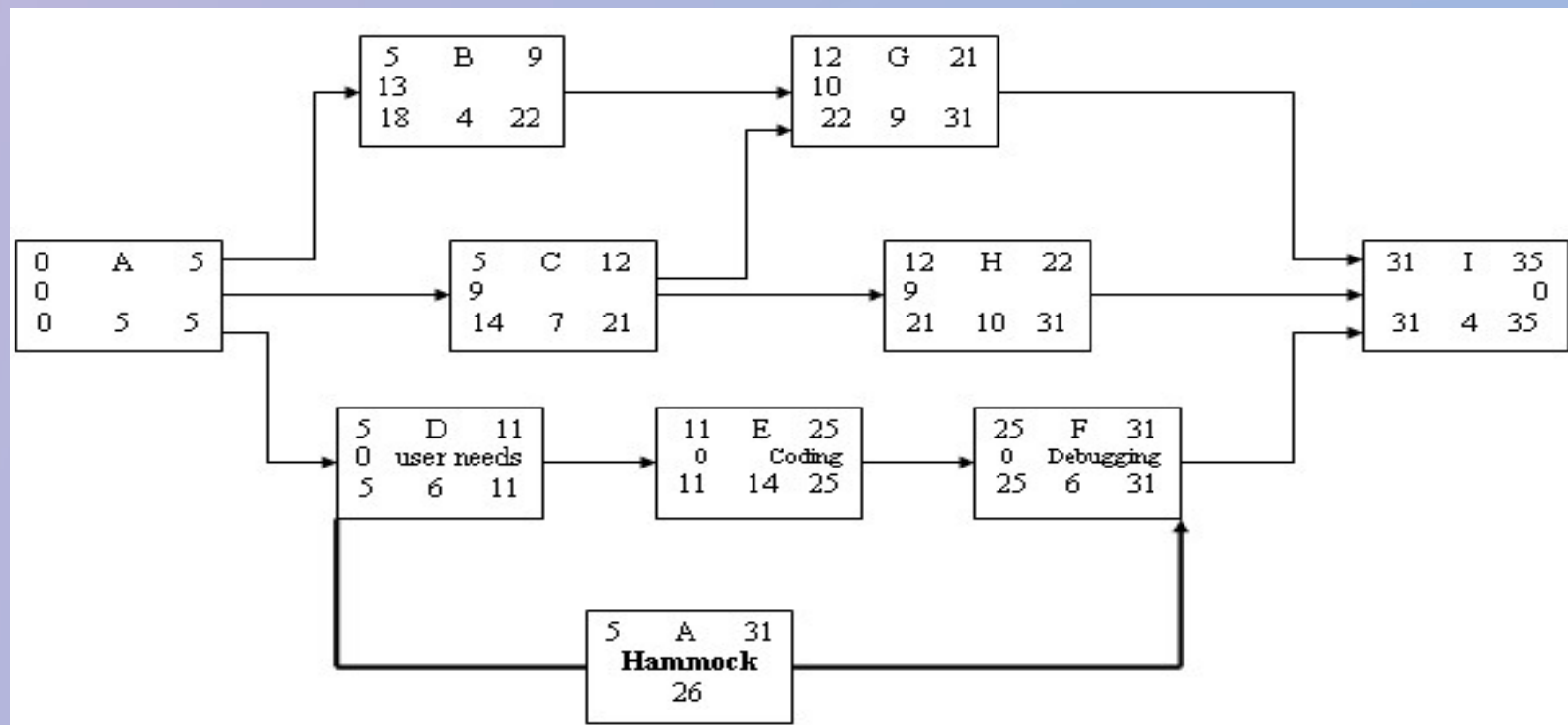
Network Activity Diagram example that Illustrates Laddering technique



If Tasks A₁, B₁ and C₁ need the **same resources** then they could run in a **laddered start**. That way once Task A₁ was completed with the resource, Task B₁ could then use it and then Task C₁.



Network Activity Diagram example that Illustrates Hammocking technique



Hammocking activities can be used as summaries for diagram segments of series activities. This allows for a smaller diagram that is less confusing.



Steps to Reduce the Critical Path

1. Eliminate tasks on the Critical Path
2. Re-plan **series** paths to be **parallel**
3. Overlap sequential tasks
4. Shorten the duration on critical path activities
 - a) Shorten earlier starting tasks
 - b) Shorten longest tasks
 - c) Shorten easiest tasks
 - d) Shorten tasks that cost the least to speed up.

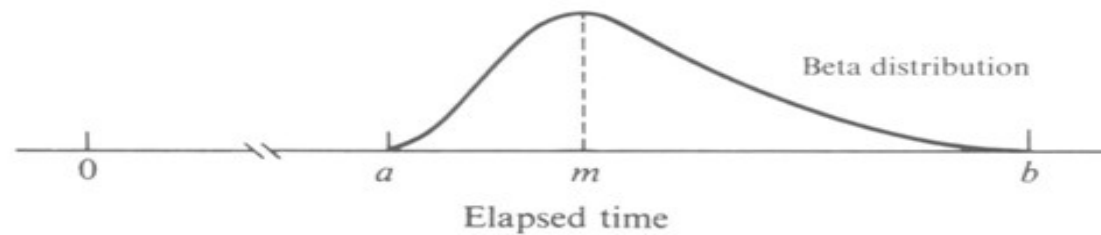


Activity Duration Estimating

- There are several methods:
- **Expert judgment estimating:** Guided by **historical information** this can be used whenever possible.
- **Analogous estimating:** A method of estimating that employs using a **similar activity** from a previous project as the basis for establishing the likely duration of a future schedule activity.
- **Parametric estimating:** The quantity of work divided by the **productivity rate**. For example, total labor hours divided by man hours per day, or cable installation in meters of cable divided by meters per hour.
- **Reserve analysis estimating:** An output of risk analysis, it can be employed to add empty buffers in project timing to deal with risk as part of a contingency plan.
- **Three point Method:** **PERT (Program Evaluation Review Technique)**



Activity Duration Estimate – **PERT Estimate**



ESTIMATED TIME FORMULA

$$\mathbf{TE = \frac{A + 4(B) + C}{6}}$$

WHERE:

A = MOST OPTIMISTIC TIME

B = MOST LIKELY TIME

C = MOST PESSIMISTIC TIME



Pert Example

- Calculate the estimated time to complete each of the following tasks.

Task	A	B	C	TE or Duration
A	12	15	17	
B	7	10	13	
C	15	19	23	
D	9	13	16	
E	24	26	30	



Pert Example

- Calculate the estimated time to complete each of the following tasks.

Task	A	B	C	TE or Duration
A	12	15	17	14.83
B	7	10	13	10
C	15	19	23	19
D	9	13	16	12.83
E	24	26	30	26.33

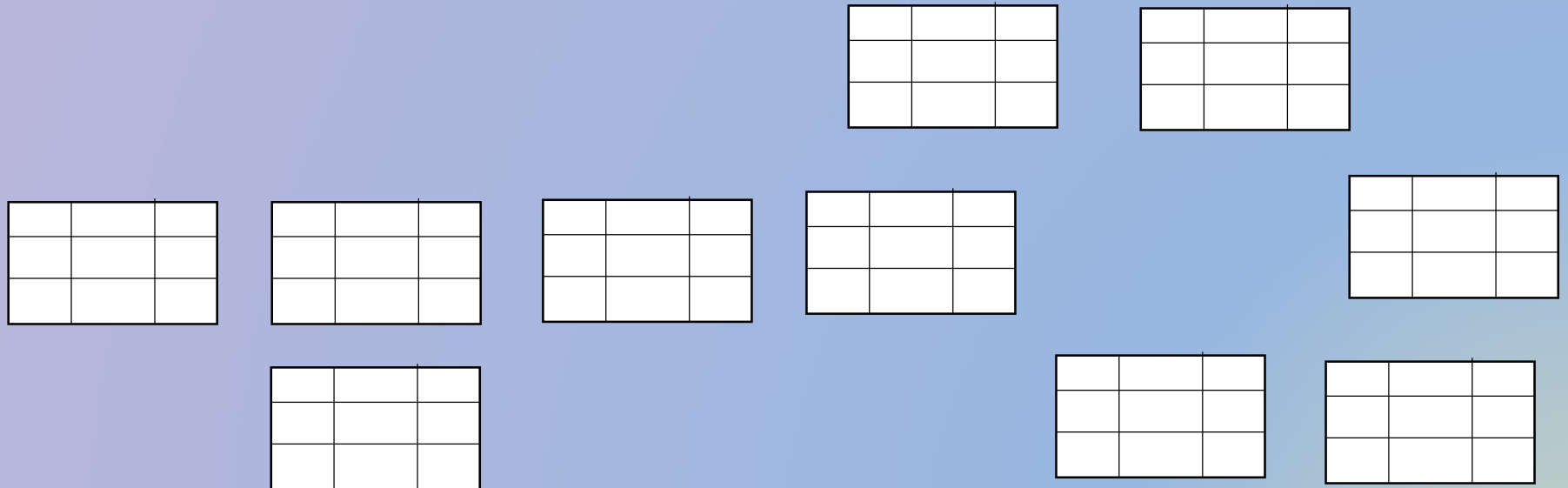


AON Example 1

Activity	Predecessors	Duration (days)
A	None	5
B	A	8
C	B	3
D	A	4
E	C	5
F	C	7
G	D, E	6
H	F	8
I	G	6
J	H, I	5

AON Example 1

Activity	Predecessors	Duration (days)
A	None	5
B	A	8
C	B	3
D	A	4
E	C	5
F	C	7
G	D, E	6
H	F	8
I	G	6
J	H, I	5



AON Example 2

Activity	Predecessors	Duration (days)
A	None	3
B	A	4
C	A	2
D	B	5
E	C	1
F	E	2
G	D, E	4
H	F, G	3
I	H	6